

■ 3.4.9 Advanced Topic: Equations Modulo Integers

When you write an equation like $lhs == rhs$, you usually want to assert that the expressions lhs and rhs are exactly equal. For some purposes, however, it is convenient to use a weaker notion of equality, and to interpret $lhs == rhs$ as meaning that two integer expressions are equal modulo some fixed integer. Solving equations that use this weaker notion of equality is important in many problems in number theory.

<code>Solve[eqns, vars, ..., Mode->Modular]</code>	solve equations with equality required only modulo an integer
<code>Modulus == p</code>	a special equation specifying the modulus used to determine equality

Solving equations modulo integers.

There are two ways to work with equations modulo integers. The first is explicitly to specify the modulus you want to use. The second is to let `Solve` find a modulus for which your equations are satisfied.

Mathematica cannot find a closed form solution for this equation over complex numbers.

```
In[1]:= Solve[5 + 11x + 17x^2 + 7x^3 + 16x^4 + x^5 == 0, x]
Out[1]= {ToRules[Roots[11 x + 17 x^2 + 7 x^3 + 16 x^4 + x^5 == -5, x]]}
```

The equation does, however, have simple solutions over integers modulo 19. Notice that if you include an equation for `Modulus`, you do not have to set `Mode->Modular` explicitly.

```
In[2]:= Solve[5 + 11x + 17x^2 + 7x^3 + 16x^4 + x^5 == 0 &&
Modulus == 19, x]
Out[2]= {{Modulus -> 19, x -> -1},
{Modulus -> 19, x -> -7}, {Modulus -> 19, x -> -12},
{Modulus -> 19, x -> -16}, {Modulus -> 19, x -> -18}}
```

If you use `Mode->Modular`, but do not explicitly include an equation for the modulus you want to use, *Mathematica* will try to find a modulus for which your set of equations can be satisfied.

The result shows that these equations are satisfied modulo two when x is odd.

```
In[3]:= Solve[{x^2 + 1 == 0, x^3 + 1 == 0}, x,
Mode->Modular]
Out[3]= {{Modulus -> 2, x -> -1}}
```

When you solve equations in the usual way over complex numbers, it takes n equations to determine the values of n complex variables. When you do not specify

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the modulus in advance, however, you can expect to get a finite number of solutions for n integer variables only by giving $n + 1$ equations. You can think of the extra equation as being what is needed to determine the modulus.

Here is a polynomial of degree three.

```
In[4]:= f = x^3 + 4x + 17
```

```
Out[4]= 17 + 4 x + x^3
```

This finds the value of x and the modulus for which both f and its derivative vanish.

```
In[5]:= Solve[ f == D[f,x] == 0, x, Mode->Modular ]
```

```
Out[5]= {{Modulus -> 8059, x -> 1001}}
```

Here are the corresponding values of f and $D[f, x]$.

```
In[6]:= {f, D[f,x]} /. %
```

```
Out[6]= {{1003007022, 3006007}}
```

They indeed do vanish modulo 8059.

```
In[7]:= Mod[%, 8059]
```

```
Out[7]= {{0, 0}}
```

Here is another example, based on a polynomial in two variables.

```
In[8]:= f = y^2 - y - x^3 + x^2
```

```
Out[8]= x^2 - x^3 - y + y^2
```

In the language of arithmetic geometry, this shows that the curve defined by $f = 0$ has a singular point modulo the prime number 11.

```
In[9]:= Solve[ f == D[f,x] == D[f,y] == 0, {x, y},  
              Mode->Modular ]
```

```
Out[9]= {{Modulus -> 11, y -> -5, x -> -3}}
```